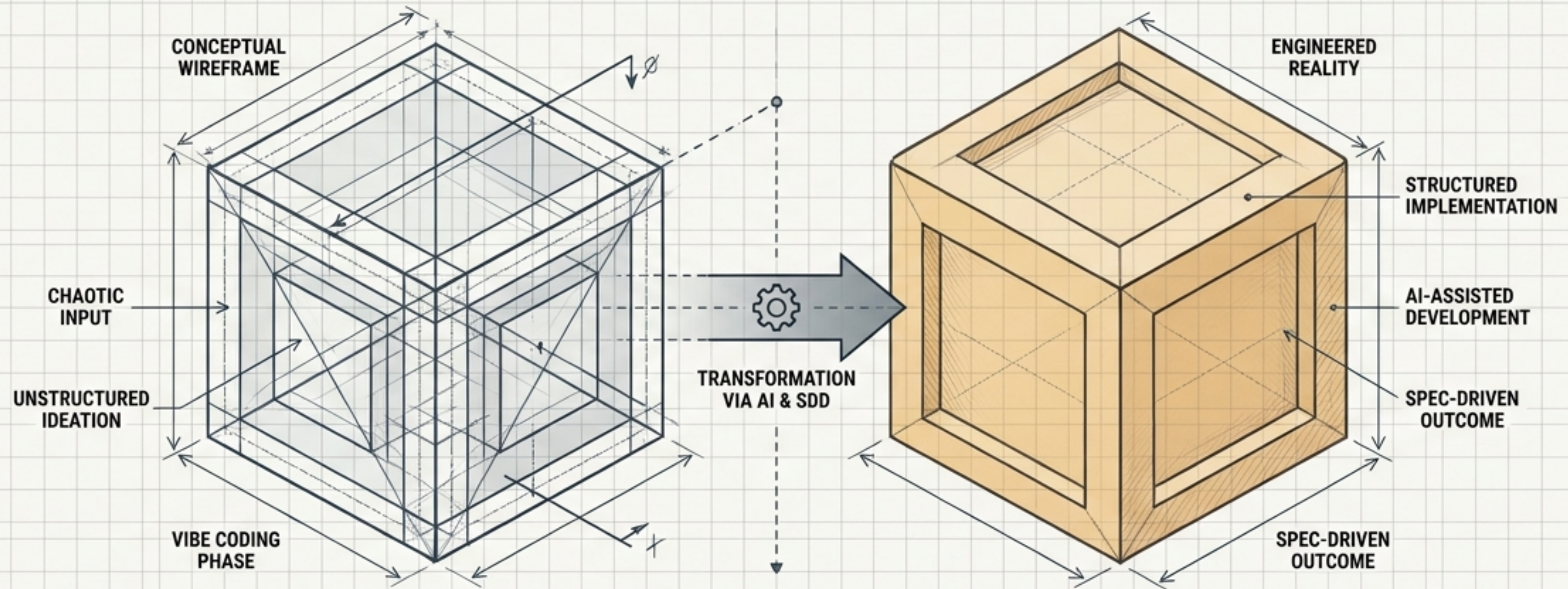


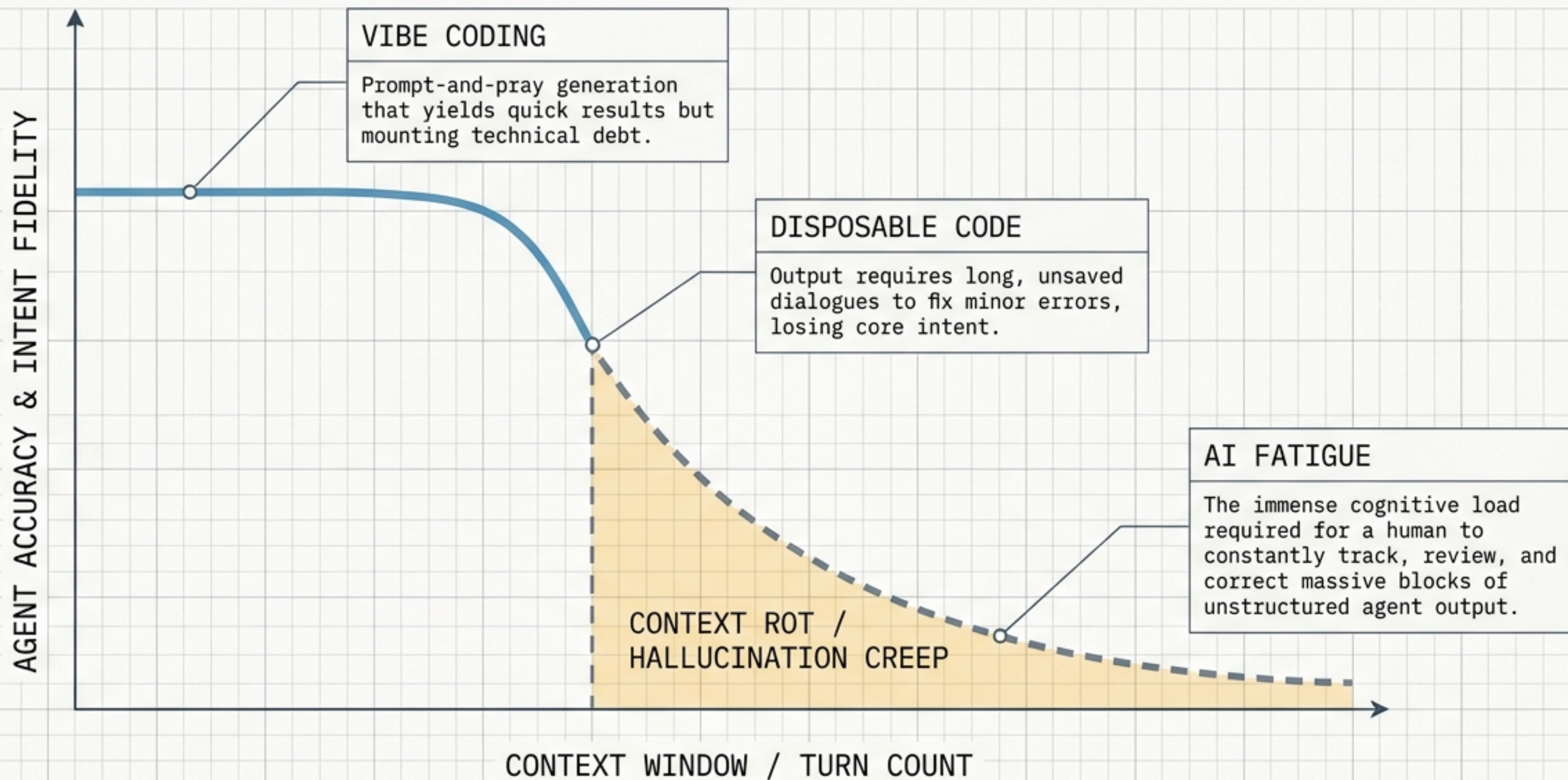
BRING BACK ENGINEERING



Moving from "vibe coding" to engineered software with AI.

A comprehensive framework for Spec-Driven Development (SDD) with coding agents

THE MOUNTING COST OF "VIBE CODING"



Decoupling the "what" from the "how"

	Vibe Coding	Spec-Driven Development
The Goal	Disposable, quick code.	Permanent, versioned technical artifacts.
The Process	Long, unrecorded dialogues.	Decoupled specifications isolated from execution.
Context Memory	Degrades as context window fills.	Preserved in immutable Markdown files.
Human Role	Exhausted vibe-checker.	Systems Architect.

THE HUMAN IS THE ARCHITECT, THE AGENT IS THE MUSCLE



THE ARCHITECT (HUMAN)

Provides the 'What' and 'Why'.

Defines the problem space, success criteria, constraints, user flows, and unique business context.

THE MUSCLE (AGENT)

Executes the 'How'.

Operates at extreme speed to write code, generate boilerplate, and implement features strictly based on the provided spec.

THE CONTRACT

The written spec is the binding agreement between human intent and agent execution. It forces alignment before a single line of code is written.

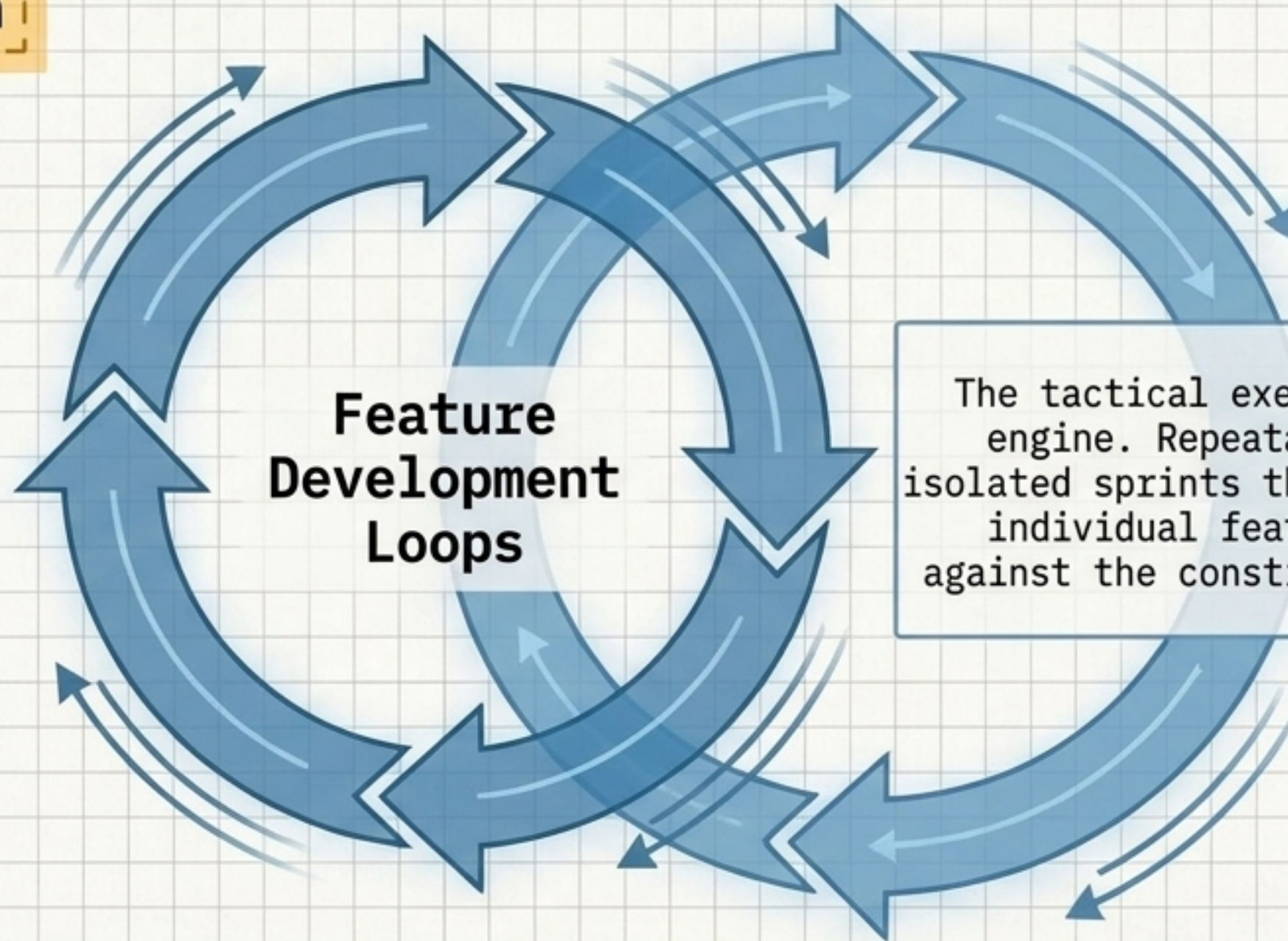
The Spec-Driven Development architecture

Project Constitution

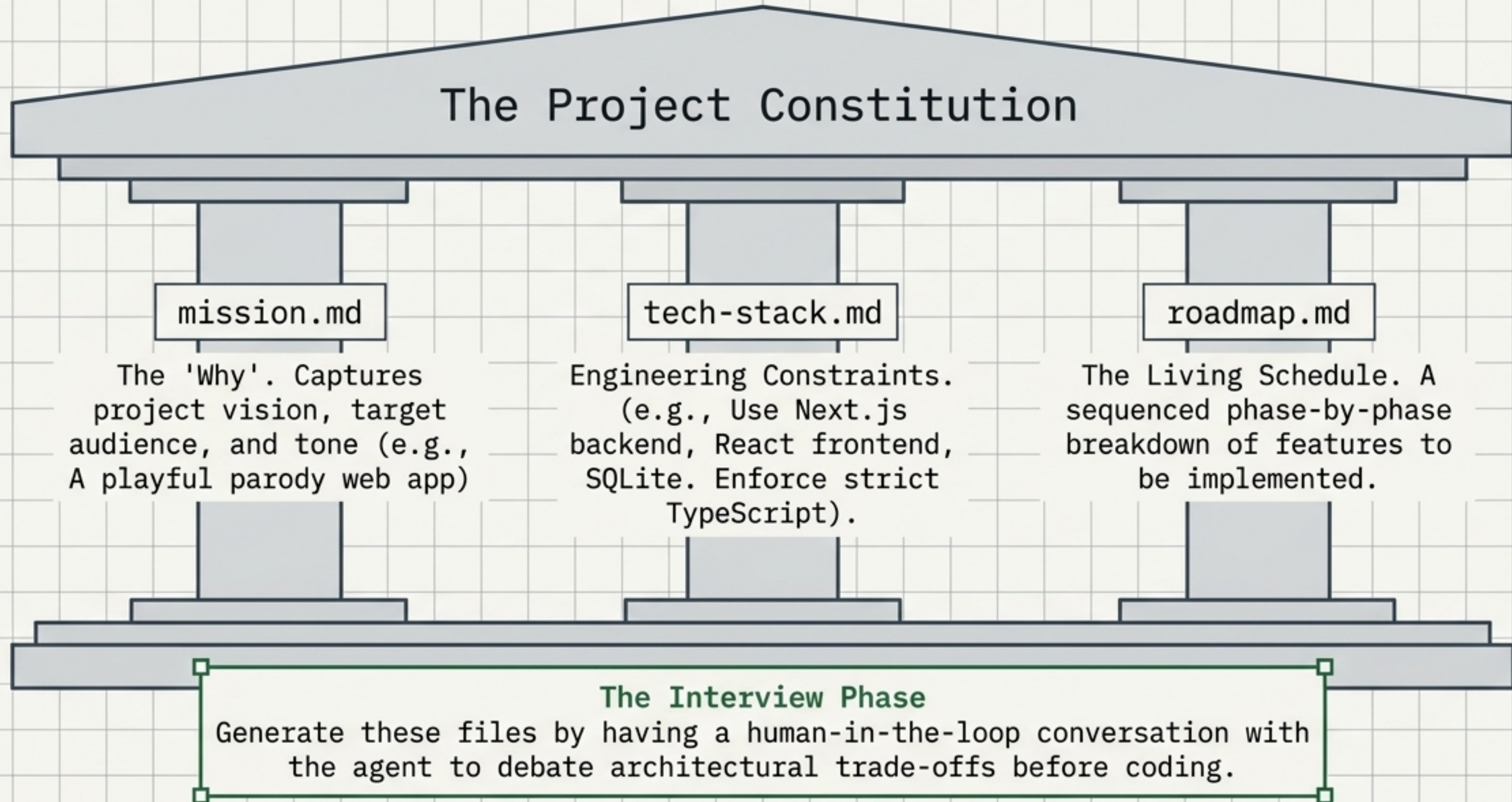
The persistent, global alignment layer.
Dictates the non-negotiables for the entire codebase.

Feature Development Loops

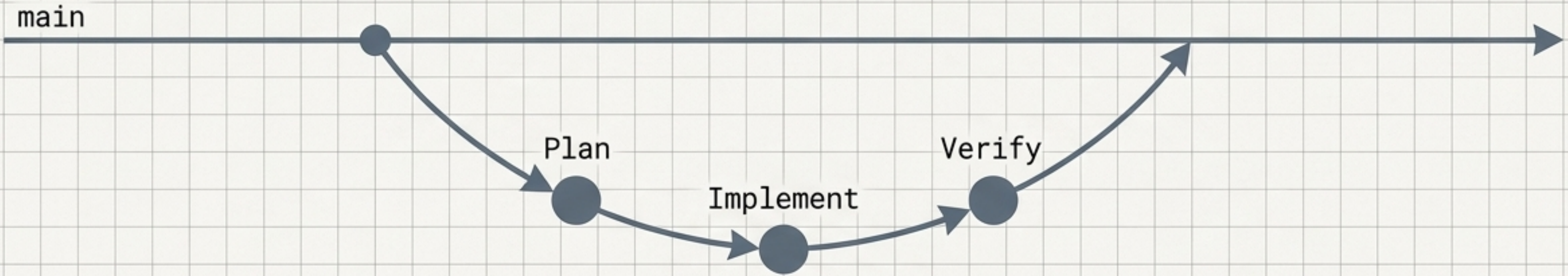
The tactical execution engine. Repeatable, isolated sprints that build individual features against the constitution.



Establishing the global rules of engagement



Isolated execution for every feature



1. Plan

Chat with the agent on a new feature branch to draft a plan, strict requirements, and a validation scorecard.

plan.md

requirements.md

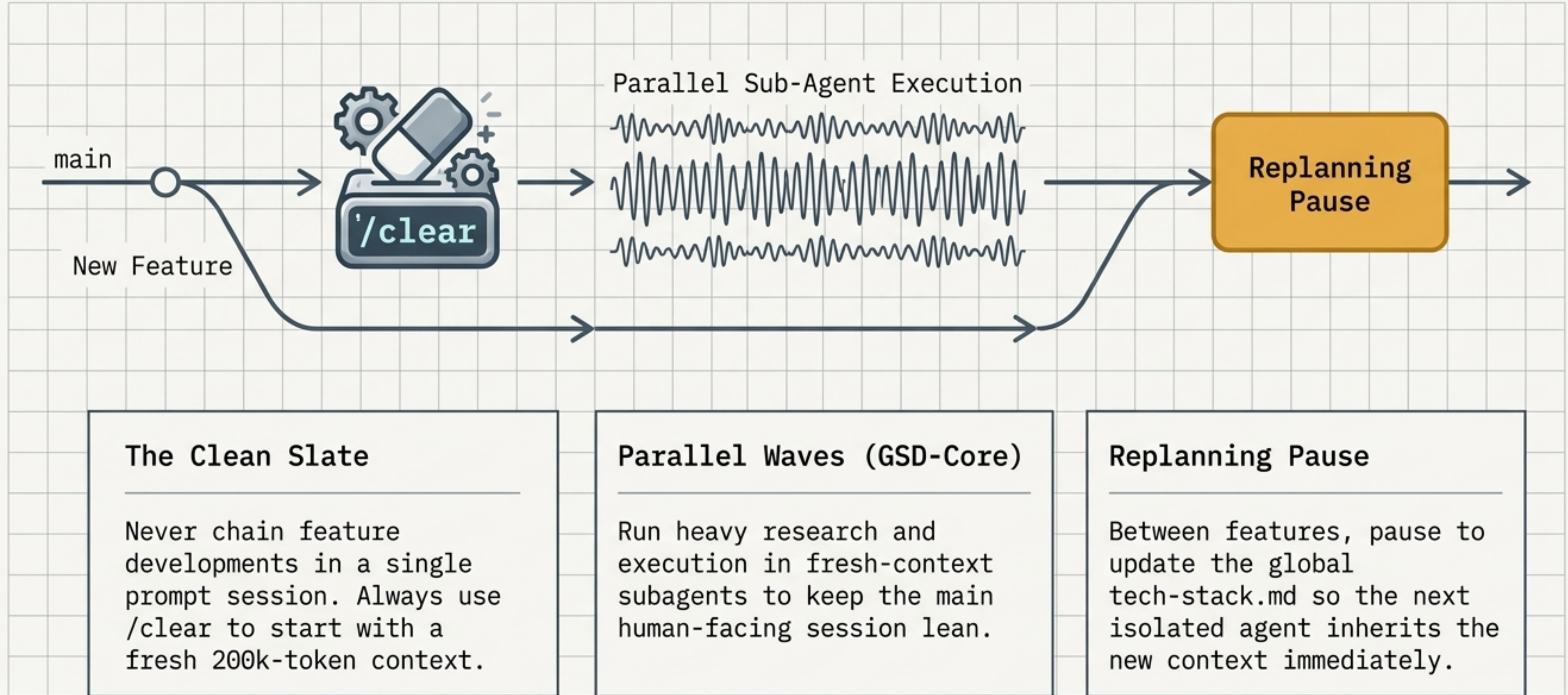
2. Implement

The agent generates the code based only on the spec. Heavy execution happens entirely autonomously.

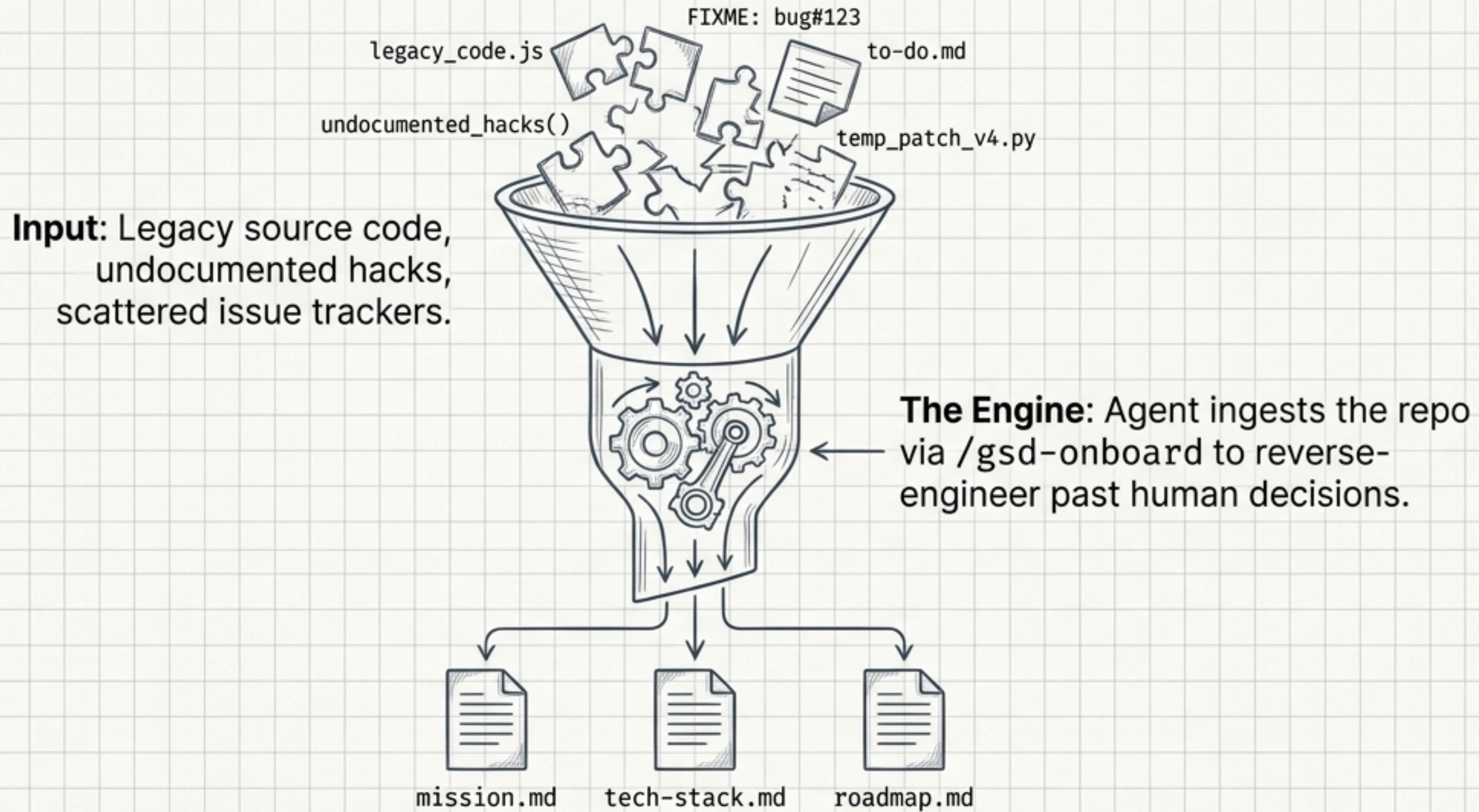
3. Verify

Human review of the contract (did it meet the spec?), running debugger tests, and spawning sub-agents for deep code review.

Stopping hallucination creep with context isolation

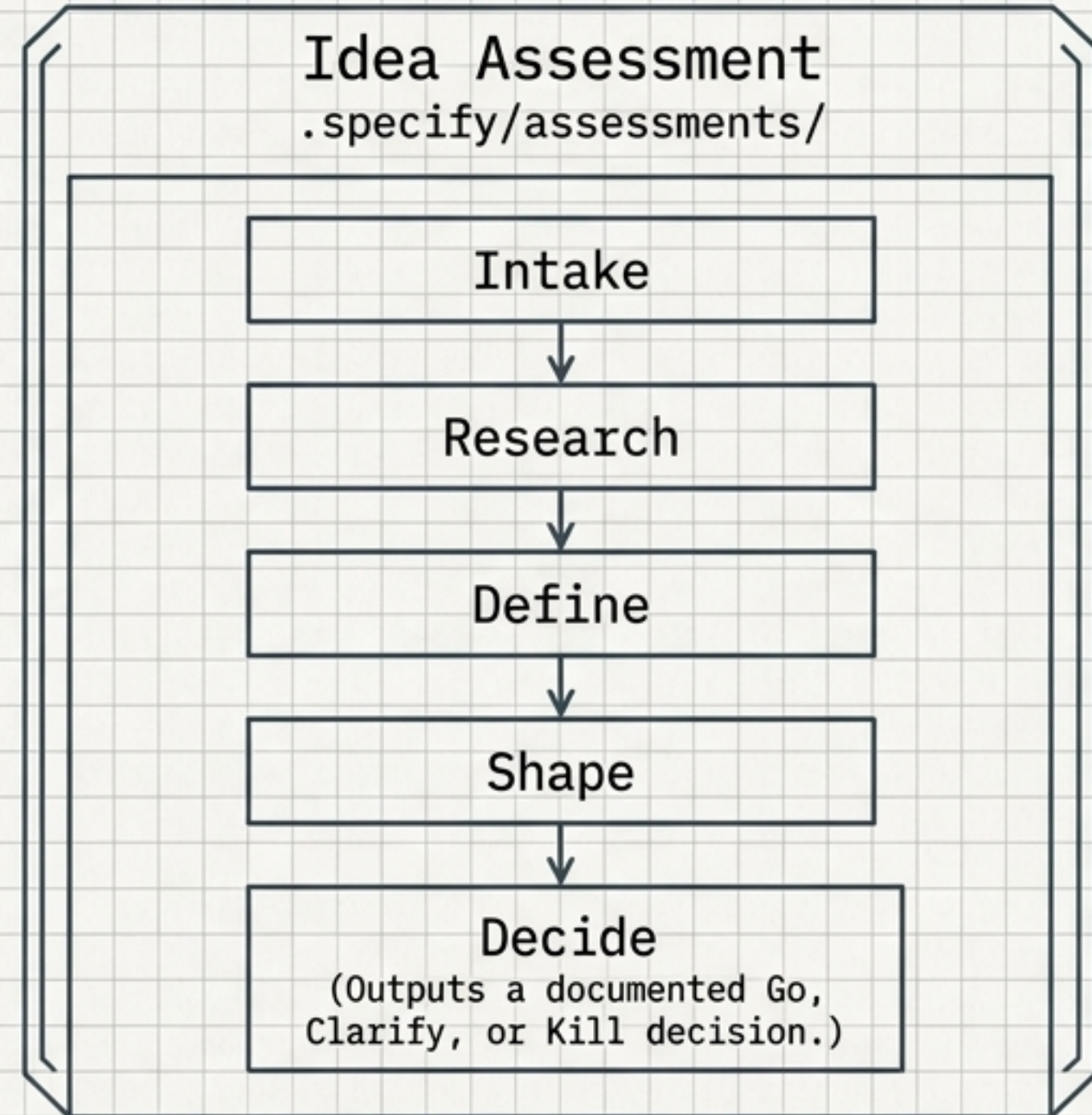
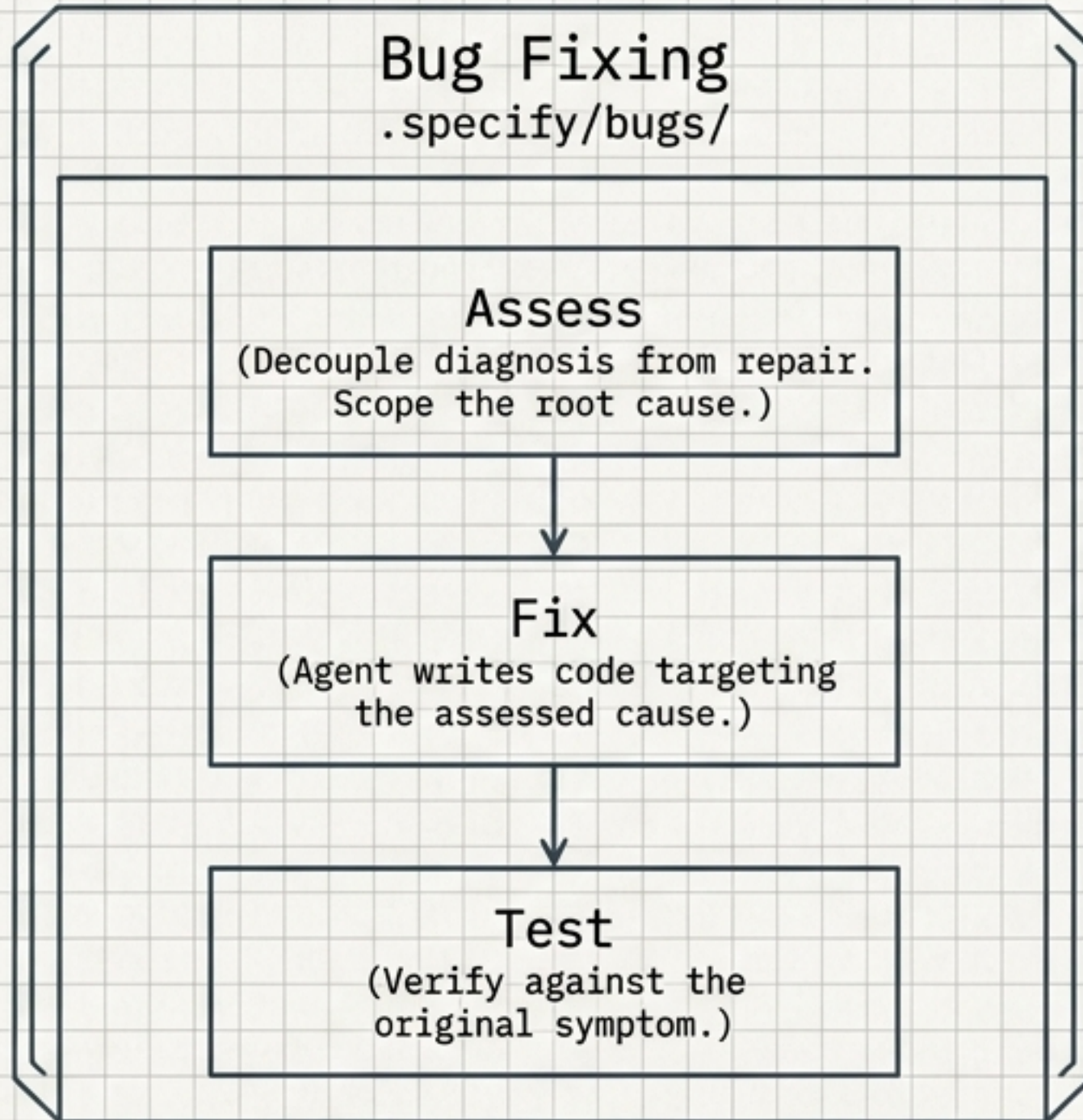


Reverse-engineering a retroactive constitution



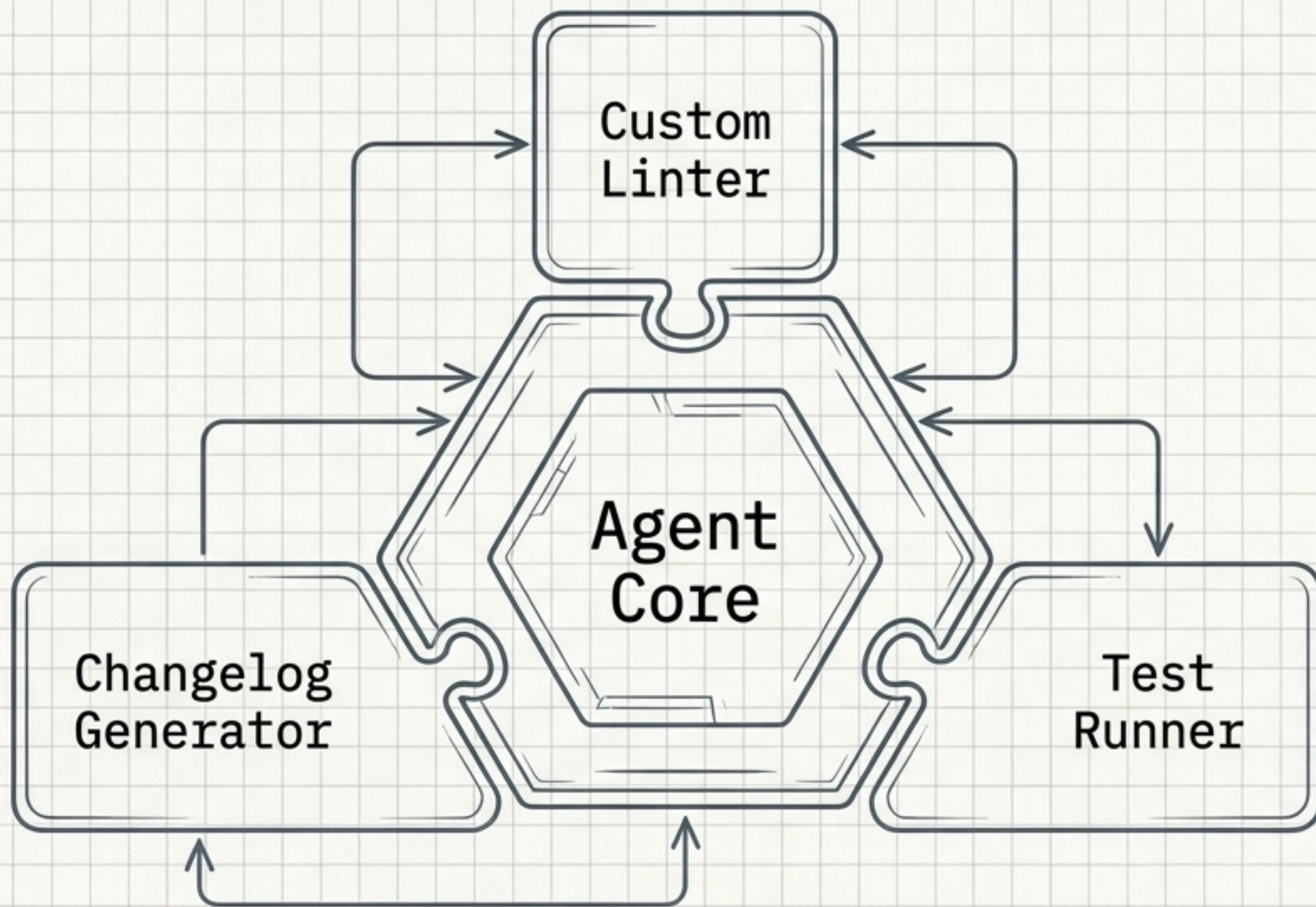
Takeaway: SDD isn't just for greenfield. This pipeline aligns all future autonomous AI code changes with the legacy decisions made by past human developers.

Expanding the methodology beyond new features



Use agents for evidence-gathering and diagnostic scoping before writing any code.

Automating repeatable workflows with Skills



Agent Skills

Packages of instructions and resources providing the agent with new, custom capabilities.

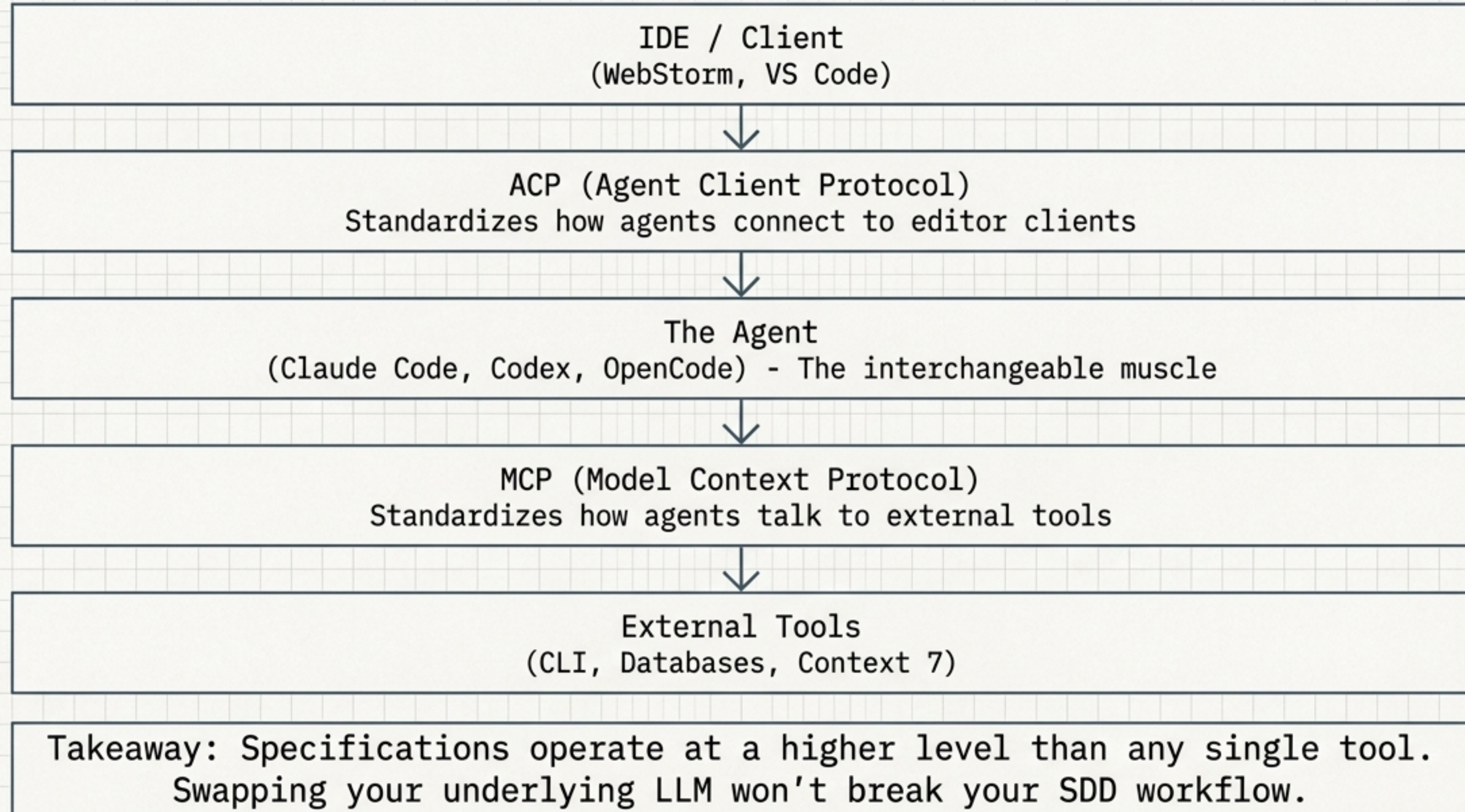
Modular Automation

Stop typing the same setup prompts. Equip skills to automate project-specific chores.

Superpowers Framework

Move from manual prompting to repeatable, context-specific workflows tailored to your organization.

Open standards prevent vendor lock-in



The open-source SDD ecosystem

Toolkits

**GitHub Spec-Kit /
OpenSpec**

Provides the entry points, folder structures, and template processes for SDD, bug fixing, and idea assessment.

Frameworks

**GSD-Core &
Superpowers**

The runtime engines. GSD handles large-scale context isolation and phase loops; Superpowers manages agentic skill execution.

Implementations

**DeepLearning.AI
AgentClinic**

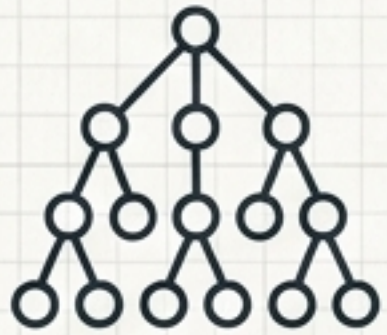
Real-world, step-by-step example repositories showing MVP builds and legacy support in action.

Taking back the driver's seat



Intent Fidelity

The AI builds exactly what you actually meant, ending the era of unpredictable generation.



Downstream Amplification

1 line of specification change cascades into 100s of lines of accurate, tested code.



Zero Context Decay

Your architectural decisions become permanent technical artifacts (STATE.md, CONTEXT.md) that survive agent memory limits.

The best code starts with a great spec. Bring back engineering.